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ORIGINAL ARTICLE

Validation of real-time viewing distance and ambient illumination measurement with two wearable sensors

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Abstract

Purpose: To report the repeatability and accuracy of viewing distance and ambient illumination measured with the Vivior and Clouclip wearable sensors in a controlled laboratory setting and while young adults performed different near visual tasks.

Methods: For Study 1, viewing distance and ambient illumination were measured repeatedly for 2 days with two sensors mounted on a mannequin. Results were compared with standard measures (ruler, lux meter) using linear regression and Bland–Altman analysis. In Study 2, viewing distances were captured in 13 young adults while they performed four near tasks for 20 min each (reading a printed article, reading an article on an iPad, typing on a laptop or watching videos on an iPad) under three illuminations (500, 250 and 25 lux). Results were compared between the devices and with a standard ruler and differences tested statistically using ANOVA and Tukey's multiple comparison tests.

Results: In Study 1, a positive linear relationship was observed between viewing distances measured by each sensor and the standard ruler (mean [SD] difference of −0.12 cm [0.23] for Vivior, −0.18 cm [0.51] for Clouclip), with good repeatability (mean [SD] difference between 2 days was 0.02 cm [0.24] for Vivior, −0.14 cm [0.35] for Clouclip). A positive linear relationship was also observed between illumination measured by each sensor and the luxmeter (mean [SD] difference of –16.85 lux [325.5] for Vivior, 39.12 lux [335.6] for Clouclip). In Study 2, the viewing distances during the four near tasks were not significantly different between the three methods used. Shorter working distances were measured with all methods when reading a printed article compared to typing on a laptop and watching a video on the iPad (all $p < 0.05$).

Conclusion: Vivior and Clouclip are recommended for real-time capture of near vision duration, distance and light exposure in future studies aiming to understand the effect of habitual visual environment.

KEYWORDS
ambient illumination, near vision behaviour, real-time viewing distance, wearable sensors

INTRODUCTION

Various factors have been investigated for their association with myopia development and progression.[1] Studies to date have found prolonged near activities with short viewing distances[2] and reduced time outdoors[3] to be associated with myopia development. However, the causative role of these two risk factors, particularly near work, remains unclear as contrary findings have been reported. For instance, the risk of myopia development in Australian

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children was associated with reading distances <30 cm and continuous reading for >30 min.[2] In contrast, a study with Singaporean children found that near work was not associated with early-onset myopia.[4] A potential source of variability in previous findings is the subjective nature of data collection of these studies, most notably using questionnaires to measure near-work activities.[4,5] It is well recognised that questionnaires used in this manner may be limited to recall bias, particularly when questions relate to time points long before administration of the questionnaire, since more recent events may affect current interpretation of the answers. Parental bias may also impact questionnaire outcomes.[6–8] For example, a recent study with children reported an underestimation of near viewing duration and an overestimation of time outdoors with parent-reported questionnaires compared to the objective estimation.[9] The potential impact of these significant biases on the accuracy of data from questionnaires has been an impetus for the development of wearable sensors that provide objective real-time quantification of near-work activities.[9]

Wearable sensors such as the Clouclip[9] (clouclip.com) and Vivior[10] (swissvisiongroup.com) devices are attached to spectacle frames and have been designed to track near-work activities in real time (i.e., duration and breaks during near work and viewing distance). These devices also monitor ambient illumination which provides information about time spent indoors or outdoors. Wearable sensors are capable of recording the frequency and duration of outdoor and indoor illumination exposure throughout the day for several days.[9] A previous study reported good repeatability of viewing distance measured with the Clouclip[11] and high correlation between ambient illumination measured by the Clouclip and a standard Lux meter.[11,12] The Clouclip has been recently used in myopia research for real-time quantification of near activities and ambient illumination exposure in children[9,13,14] and adults.[15]

In recent years, alternatives to the Clouclip have become available to measure near-work activities. One such device is the Vivior, which was initially developed to monitor visual behaviour after cataract or refractive surgery,[10] and has recently been used to monitor the visual behaviour of myopic children.[16] In addition to real-time near viewing distance and ambient illumination measurements, the Vivior is capable of quantifying other factors such as head motion and exposure to different wavelength spectra, thereby expanding the application of this device in myopia research. The validation of viewing distance and ambient illumination measurements with the Vivior has not yet been reported. Additionally, to the best of our knowledge, no studies have compared real-time viewing distance measurements between the two wearable sensors. To address these limitations in knowledge, this study had two aims: (a) to report the repeatability and reliability of viewing distance and ambient illumination measurements with the Vivior and Clouclip in a controlled laboratory setting and (b) to compare the accuracy of viewing distance measured simultaneously by Vivior and Clouclip sensors in young adults while they performed different near tasks.

Key points

- Vivior and Clouclip sensors accurately measured viewing distances. However, data loss has been reported with the Clouclip when it enters sleep mode due to lack of movement.
- Vivior and Clouclip sensors were able to differentiate between outdoor (>1000 lux) and indoor (<1000 lux) illuminations. However, >1000 lux illumination was recorded by Vivior and Clouclip for some indoor locations facing bright sunlight.
- Variable ambient illumination was recorded by Vivior and Clouclip sensors compared to the standard luxmeter for outdoor illumination exceeding 3500 lux.

METHODS

Study design

Two studies were conducted to address the two aims of this research. Study 1 evaluated the repeatability of viewing distance and ambient illumination captured with the two wearable sensors using a mannequin head and compared it to standard measurements. Study 2 compared the accuracy of viewing distance measurement between the two wearable sensors in young adults while they performed different near tasks.

Study 1: Validation of viewing distance and ambient illumination

Each of the five Vivior and five Clouclip sensors was individually attached to the right side of a spectacle frame using an adaptor clip or rubber sleeve, respectively, and placed on a mannequin head (Figure 1). The mannequin head was positioned at varying distances from a desktop monitor (Dell, 12.6 [H] × 21.3 [W] inch, dell.com) in 5 cm increments between 10 and 100 cm. Viewing distances to the monitor were measured using all 10 devices on two consecutive days to assess for repeatability. Viewing distances were also measured with a millimetre ruler placed at the same height as the device on the mannequin head to evaluate the accuracy of the measurements with the two devices. The millimetre ruler position was constantly monitored to minimise parallax error. Viewing distances were measured at each testing position for 5 min and under standard photopic indoor room illumination (400–500 lux).



FIGURE 1 Experimental set-up of viewing distance and ambient illumination measurements with the Vivior device attached to a spectacle frame placed on a mannequin head in Study 1.

Ambient illumination was measured with Vivior and Clouclip sensors attached to the spectacle frame placed on the mannequin head (Figure 1), as described above, on three different days at seven different indoor and outdoor locations at the University of New South Wales (UNSW) Sydney, Australia, Kensington campus: outdoor area under direct sunlight, outdoor area under the shade, indoor meeting room, indoor corridor, indoor clinical examination room and indoor entrance to a building and near a glass window.[11,12] To assess the reliability of the sensors, ambient illumination was simultaneously measured with a lux meter (Chroma Meter CL-200A, Konica Minolta Inc., sensing.konicaminolta.us) that was placed at the same level and direction as the illumination sensors on the Vivior and Clouclip devices. The lux meter was calibrated at the UNSW Sydney Optics and Radiometry laboratory before data collection. The mannequin head was placed at each location for 12 min and illumination was recorded by the luxmeter six times at 2-min intervals and for the entire duration with the Vivior and Clouclip sensors. Ambient illumination was measured from all five Vivior and Clouclip sensors on the same day for each location within a period of 30 min.

During all measurements, the mannequin head was gently tilted side by side by the investigator for less than 5 s without touching the sensors every 5 min and 40 s[11] for the Vivior and Clouclip, respectively, to prevent the sensors from going into sleep mode and the possibility of losing data.

Study 2: Viewing distance measurement accuracy with wearable sensors during near work

This study was approved by the UNSW Human Research Ethics Committee (HC230396) and adhered to the tenets of the Declaration of Helsinki. During each of the three study visits, participants were asked to wear Vivior and Clouclip sensors attached to their habitual spectacle frames or were provided with spectacle frames with plano lenses if they required no refractive error correction or were contact lens wearers and performed four different 20-min duration near tasks in random order at their habitual viewing distance (Figure 2). The tasks included reading a printed article (paper size 30.5 cm × 22 cm and approximately 11 font size), reading an article on an Apple iPad (apple.com; 9th generation, 10.2-inch screen and approximately 11 font size), typing on a laptop computer (14-inch screen) (Dell, Latitude 5420, dell.com/en-us) and watching videos on the iPad. During all four near tasks, the Clouclip was manually tilted 10° downward to support the capture of viewing distance.[11] Viewing distances were also measured every 2 min with a millimetre ruler placed at the same level and angle of direction as the Vivior and Clouclip sensors. The viewing distance during each near task was measured under three illumination levels: 500 lux (standard photopic room illumination), 250 and 25 lux (mesopic condition), randomly assigned for each of the three study visits. The illumination level remained fixed through each study visit. The three light levels were selected as indoor photopic illumination typically varies between 30 and 500 lux.[17] To avoid the sensors going into sleep mode, the spectacle frame was gently shaken by the investigator every 40 s for less than 5 s without touching the sensor. Participants

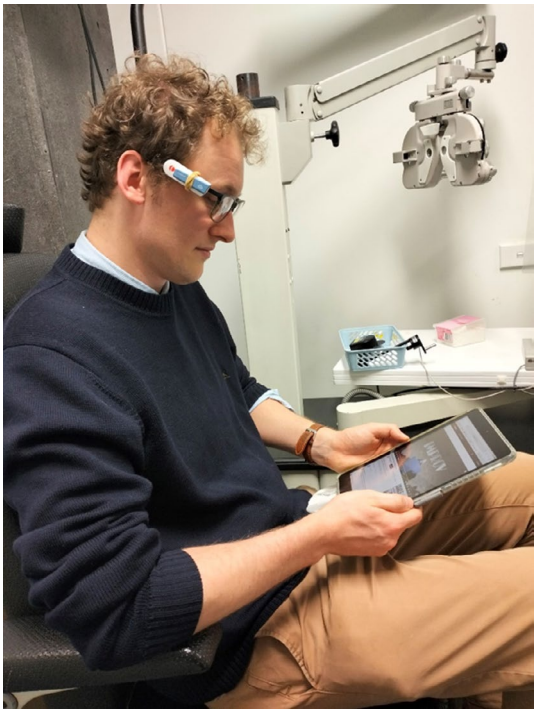


FIGURE 2 Experimental set-up of viewing distance capture with the Vivior and Clouclip attached to a spectacle frame worn by a participant (consent given) watching videos on an iPad in Study 2.

were informed before this was done and that it was to avoid losing data.

The Clouclip goes into sleep mode if no movement is detected for more than 40 s,[11] while for the Vivior, the threshold time before sleep mode was 5 min. Participants are not expected to move their heads every 40 s. Therefore, to avoid sensors going into sleep mode and potential data loss, spectacles which had both sensors attached were gently shaken every 40 s.

Participants

A total of 13 healthy young adult participants aged between 18 and 35 years with best-corrected monocular distance visual acuity of 0.10 LogMAR or better and without ocular pathology were recruited. To confirm eligibility, participants were invited for a screening visit that included measurements of visual acuity using a computerised vision chart (Test Chart 2000 Pro, thomson-software-solutions.com/), objective refraction using the Shin-Nippon NVision K-5001 open-field autorefractor (rexxam.co.jp/eye-care/), subjective refraction using standard optometric techniques, as well as anterior and posterior eye health examinations using a slit-lamp biomicroscope. Following the screening visit, subjects were scheduled for three study visits of up to 80-min duration each with a minimum 2-day gap between the visits. All participants completed the three study visits within 7–10 days.

Wearable sensors

Five Vivior Mark 3+ and five second-generation Clouclip devices were used in both studies. The Vivior Mark 3+ device is a 14-g USB-like device (dimensions: 60 × 19 × 11 mm) with three sensors. Two sensors measure viewing distance with one sensor positioned in the primary gaze direction and the other tilted 30° downward from the horizontal meridian. The additional sensor measures viewing distance during near activities conducted during downgaze without head tilt. The Vivior can track viewing distances between 10 and 200 cm. The third is an illuminance sensor designed to measure ambient illumination within the central ±20° visual field (Figure 3). The standard sampling rate of viewing distance and ambient illumination measurements with the Vivior is 5 min, and a customised 1-min sampling rate was used in these studies. The Vivior can capture illumination levels between ≤1 and ≥100,000 lux as observed during this study.

The Clouclip is a 4.7-g USB-like device (dimensions 45.3 × 13.4 × 8.0 mm) with two sensors, one for capturing viewing distance and another for ambient illumination (Figure 3). The Clouclip has a fixed inbuilt sampling rate of measuring viewing distance every 5 s and ambient illumination every 2 min. Clouclip uses a tracking beam with a 25° diameter that is directed 10° down from the horizontal meridian to detect viewing distances between 5 and 120 cm and capture illumination levels between 1 and 655,336 lux.[13,14]

Data and statistical analysis

Immediately after data collection, Vivior and Clouclip sensors were synced to their respective apps and data uploaded to an online cloud provided by each manufacturer. For the Vivior, raw data downloaded from the online cloud were provided in a CSV spreadsheet by the manufacturer who was masked from the subject and study protocol. For Clouclip, raw data were downloaded from the Cloud into a CSV spreadsheet that was directly accessible to study investigators. The Vivior and Clouclip captured the outcome parameters as a singular measurement at the respective frequencies.

For Study 1, the mean values of viewing distances captured by both sensors and ruler for 5 min at each testing distance (10–100 cm) and illumination captured by the sensors and luxmeter for 12 min at each testing location were calculated. For Study 2, the mean values of viewing distances in young adults captured by the sensors and standard ruler during each of the four near tasks were

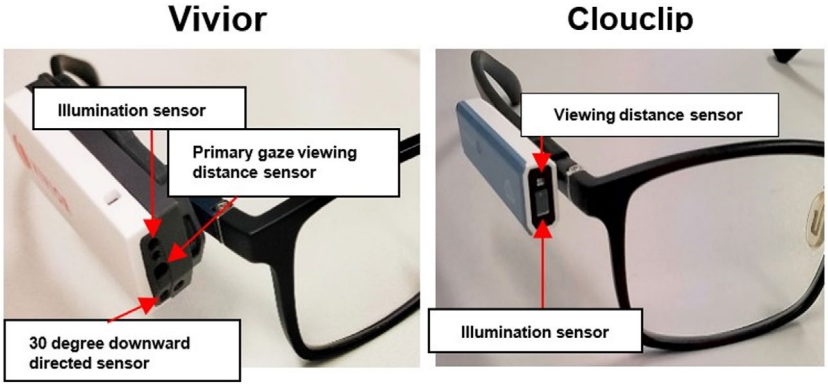


FIGURE 3 Functional parts of the two wearable sensors, the Vivior (left) and Clouclip (right).

calculated. Although the spectacle frame was shaken every 40 s to prevent the Clouclip from going into sleep mode, there was missing data for about 1–2 min during each near task . Therefore, for the Clouclip, mean values were calculated for viewing distances captured for 18 min instead of 20 min during each of the four near tasks.

Statistical analysis was performed using GraphPad Prism version 9.0 for Windows (graphpad.com) and SPSS (v27, ibm.com) software. All data followed a normal distribution. For Study 1, simple linear regression analysis was performed to detect any relationship between the viewing distance captured by the wearable sensors and standard ruler and between ambient illumination measured by the wearable sensors and the luxmeter. Then, a Bland–Altman analysis was performed to assess the repeatability of viewing distance and ambient illumination captured by the Vivior and Clouclip sensors. One sample *t*-test was performed to determine differences in the viewing distance-measurements between the Vivior, Clouclip and standard ruler, as well as ambient illumination measurement between the Vivior, Clouclip and lux meter. Two-way mixed, average measures and absolute agreement models were used to determine the inter-device intraclass correlation coefficient (ICC) for viewing distance measurements with the five Vivior and five Clouclip sensors. Fisher's exact test was performed to check the sensitivity and specificity of Vivior, Clouclip and Luxmeter to capture the illumination at outdoor and indoor locations using the >1000 lux cut-off.[18] For Study 2, two-way ANOVA and Tukey's multiple comparison tests were performed to determine differences in the viewing distances between different near tasks and with the three measurement types, standard ruler, the Vivior and the Clouclip.

RESULTS

Study 1

Validation of viewing distance measurements

A positive linear relationship was observed between the viewing distance measured by the Vivior and standard ruler ($y = 1.00x + 0.11$, $R^2 = 0.99$, $p < 0.001$) and the Clouclip and the standard ruler ($y = 1.00x - 0.02$, $R^2 = 0.99$, $p < 0.001$). Bland–Altman analysis showed good repeatability in viewing distance measured by the Vivior (Figure 4) and Clouclip (Figure 5) on two consecutive days. Overall, the mean (SD) differences in viewing distance measurements between the 2 days for all the five Vivior and Clouclip sensors were 0.02 cm (0.24) [limits of agreement, LOA −0.45 to +0.50 cm] ($t = 0.43$, df = 18, $p = 0.67$) and −0.14 cm (0.35) [LOA −0.84 to +0.55 cm] ($t = 1.75$, df = 18, $p = 0.09$), respectively, which were not statistically significant. The mean differences in viewing distance measurement between the 2 days for each of the five Vivior and Clouclip readings were also not statistically significant (Table 1). Bland–Altman analysis also revealed good agreement in viewing distance measurements between Vivior and Clouclip sensors, as well as each device with the standard ruler (Figures 4 and 5). Overall, the mean (SD) difference in viewing distance measured by the standard ruler and all five Viviors was −0.12 cm (0.23) [LOA −0.56 to +0.32 cm] ($t = 2.32$, df = 18, $p = 0.03$) and −0.18 cm (0.51) [LOA −1.17 to +0.81 cm] ($t = 1.54$, df = 18, $p = 0.14$) for the standard ruler and all five Clouclips (Table 1). The inter-device ICC for both the Vivior (0.99) and Clouclip (0.99) indicated good reliability.

Validation of ambient illumination measurements

There was a linear trend observed between the ambient illumination measured by the Vivior and lux meter ($y = 1.01x + 4.44$, $R^2 = 0.95$, $p < 0.001$) and Clouclip and lux meter ($y = 0.95x + 20.27$, $R^2 = 0.96$, $p < 0.001$; Figure 6). Bland–Altman analysis showed high agreement in the ambient illumination measurements in outdoor and indoor locations between the luxmeter, Vivior and Clouclip devices (Figure 7). The mean difference in illumination measurements between the luxmeter and Vivior was −16.85 lux (325.5) (LOA −564.8 to +545.8 lux) and 39.12 lux (335.6) (LOA −618.6 to +696.8 lux) between the luxmeter and Clouclip. One sample *t*-test showed no significant difference in illumination measurements between the luxmeter and Vivior ($t = 0.94$, df = 329, $p = 0.35$) and luxmeter and Clouclip ($t = 1.74$, df = 225, $p = 0.10$). Bland–Altman analysis revealed increased variability in illumination measurements between the luxmeter and wearable sensors at levels greater than 3500 lux (Clouclip) and 4000 lux (Vivior) (Figure 7).

For all outdoor locations, the Vivior, Clouclip and Luxmeter always captured measurements >1000 lux illuminations. However, Vivior, Clouclip and Luxmeter recorded >1000 lux illumination for some indoor locations, such as near a window or glass door on a bright sunny day. With the >1000 lux cut-off value,[18] the sensitivity values of detecting illumination at outdoor locations for Vivior, Clouclip and Luxmeter were 80.33%, 85.44% and 86.27%, respectively. The specificity of detecting indoor illumination using the ≤1000 lux criteria for Vivior, Clouclip and Luxmeter was 100%.

Study 2

A power calculation was performed to determine whether the small sample size ($n = 13$) was sufficient to support the findings. The study power calculation with G*Power 3.1.9.7 (psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower) showed 0.83 with an alpha error probability 0.05. The mean (SEM) age of the participants (9 females and 4 males) was 31.15 (0.67) years. The mean (SEM) spherical equivalent refraction was −2.49 D (0.82) for the right eye and −2.31 D (0.78) for the left

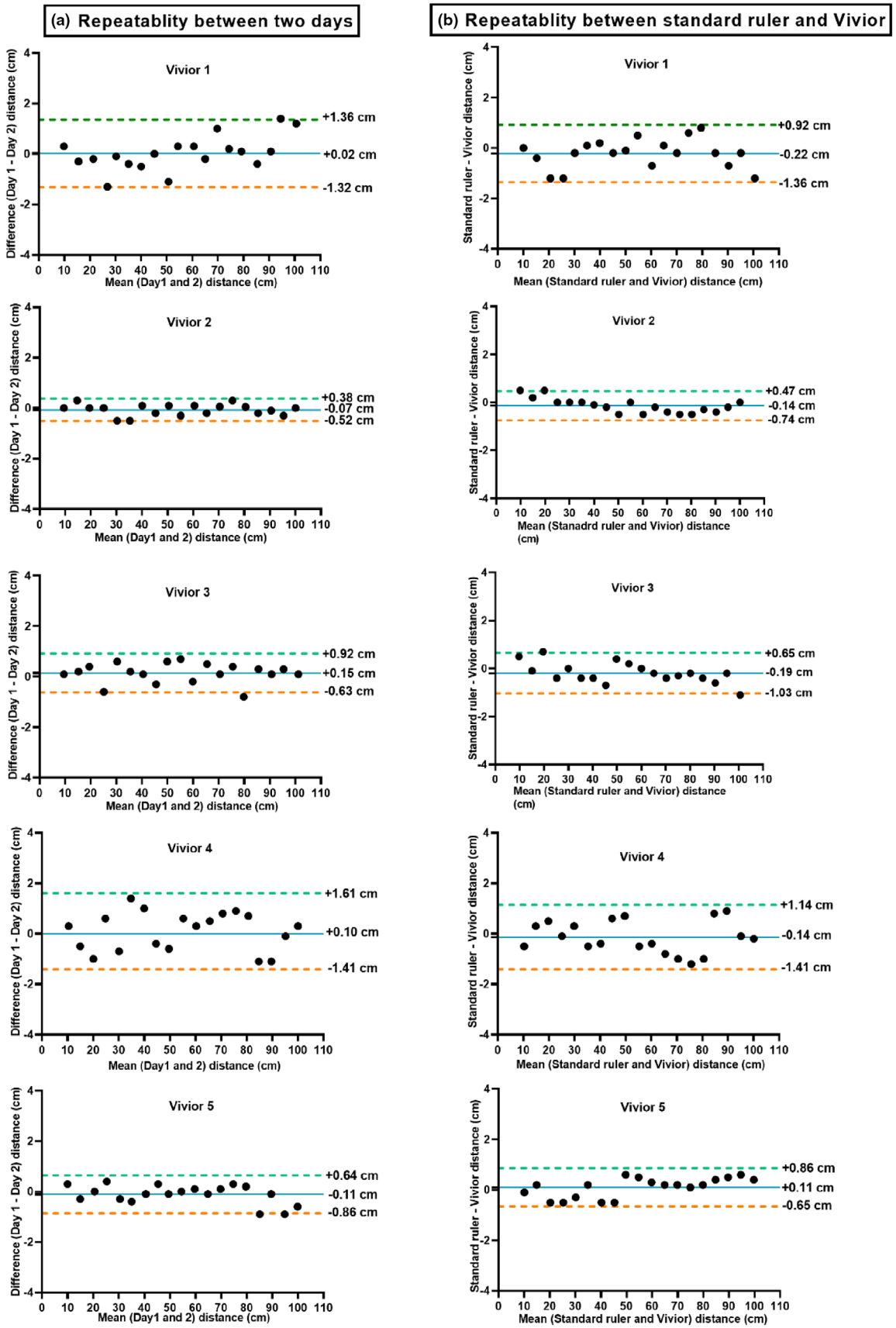


FIGURE 4 Bland–Altman plots of viewing distance measured by the five Vivior devices (a) between 2 days and (b) in comparison to the standard ruler. The dashed green and orange lines represent the upper and lower limits of agreement, respectively, and the solid blue line indicates the mean bias.

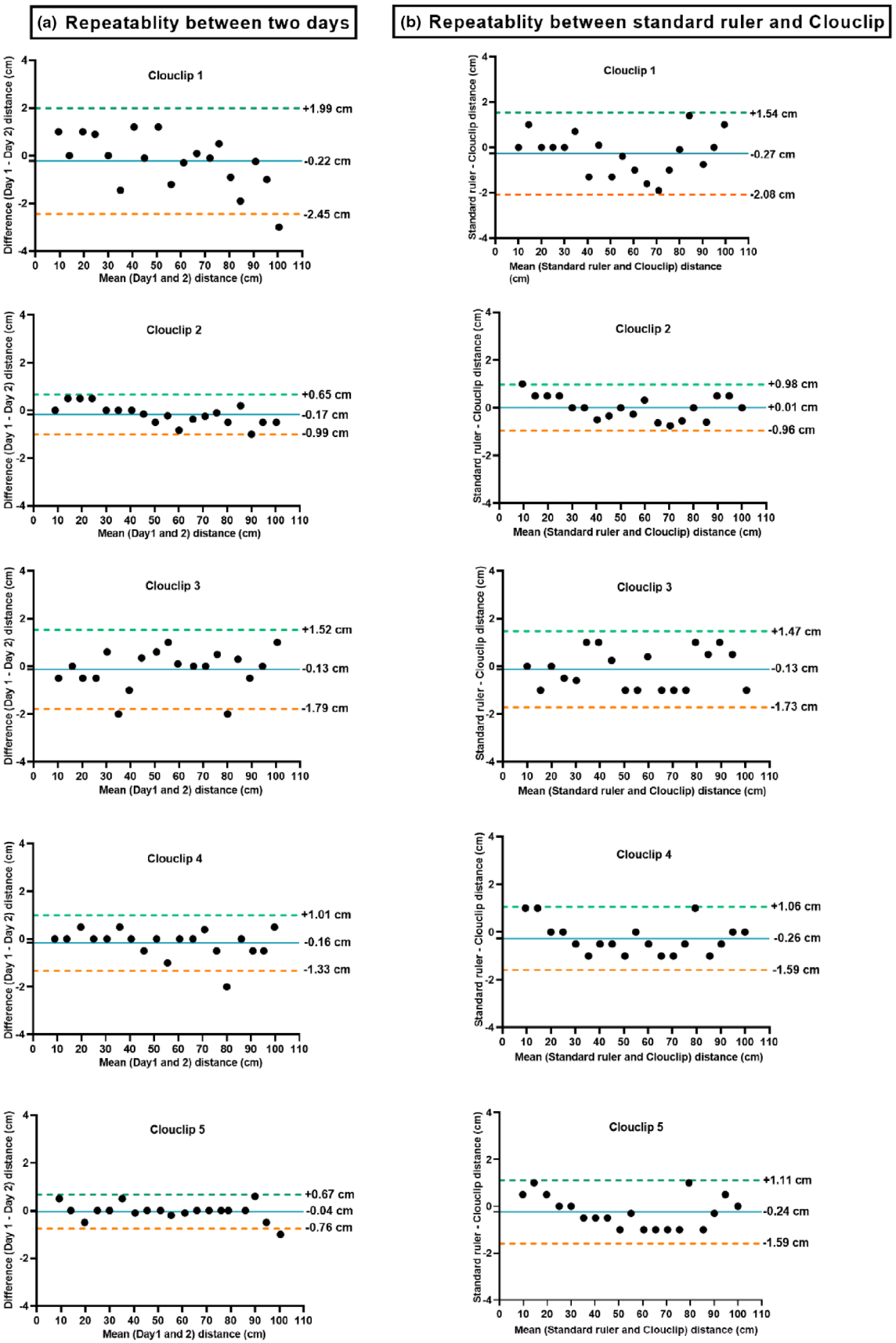


FIGURE 5 Bland–Altman plots of viewing distance measured by the five Clouclip devices (a) between 2 days and (b) in comparison to the standard ruler. The dashed green and orange lines represent the upper and lower limits of agreement, respectively, and the solid blue line indicates the mean bias.

TABLE 1 Mean (SEM) differences in viewing distance measured by the sensors on two consecutive days and between the sensors and standard ruler.

Sensor	(Day 1–Day 2) viewing distance (cm)
Vivior 1	0.02 cm (0.68) (LOA −1.32 to +1.36 cm) ($t = 0.13$, $p = 0.89$)
Vivior 2	−0.07 cm (0.22) (LOA −0.52 to +0.52 cm) ($t = 1.29$, $p = 0.21$)
Vivior 3	0.15 cm (0.39) (LOA −0.63 to +0.92 cm) ($t = 1.63$, $p = 0.12$)
Vivior 4	0.10 cm (0.77) (LOA −1.41 to +1.61 cm) ($t = 0.57$, $p = 0.58$)
Vivior 5	−0.11 cm (0.38) (LOA −0.86 to +0.64 cm) ($t = 1.25$, $p = 0.22$)
Clouclip 1	−0.22 cm (1.13) (LOA −2.45 to +1.99 cm) ($t = 0.87$, $p = 0.39$)
Clouclip 2	−0.17 cm (0.42) (LOA −0.99 to +0.65 cm) ($t = 1.76$, $p = 0.09$)
Clouclip 3	−0.13 cm (0.84) (LOA −1.79 to +1.52 cm) ($t = 0.68$, $p = 0.49$)
Clouclip 4	−0.16 cm (0.59) (LOA −1.33 to +1.01 cm) ($t = 1.19$, $p = 0.25$)
Clouclip 5	−0.04 cm (0.36) (LOA −0.76 to +0.67 cm) ($t = 0.50$, $p = 0.62$)
Sensor	**(Standard ruler–Sensor) viewing distance (cm)**
Vivior 1	−0.22 cm (0.58) (LOA −1.36 to +0.92 cm) ($t = 1.65$, $p = 0.11$)
Vivior 2	−0.14 cm (0.31) (LOA −0.74 to +0.47 cm) ($t = 1.93$, $p = 0.07$)
Vivior 3	−0.19 cm (0.43) (LOA −1.03 to +0.65 cm) ($t = 1.92$, $p = 0.07$)
Vivior 4	−0.14 cm (0.65) (LOA −1.41 to +1.14 cm) ($t = 0.92$, $p = 0.37$)
Vivior 5	0.11 cm (0.39) (LOA −0.65 to +0.86 cm) ($t = 1.18$, $p = 0.24$)
Clouclip 1	−0.27 cm (0.92) (LOA −2.08 to +1.54 cm) ($t = 1.28$, $p = 0.22$)
Clouclip 2	0.01 cm (0.49) (LOA −0.96 to +0.98 cm) ($t = 0.08$, $p = 0.93$)
Clouclip 3	−0.13 cm (0.83) (LOA −1.73 to +1.47 cm) ($t = 0.69$, $p = 0.50$)
Clouclip 4	−0.26 cm (0.67) (LOA −1.59 to +1.06 cm) ($t = 1.70$, $p = 0.10$)
Clouclip 5	−0.24 cm (0.68) (LOA-1.59 to +1.11 cm) ($t = 1.53$, $p = 0.13$)

Abbreviations: cm, centimetre; LOA, limits of agreement; SEM, standard error of mean.

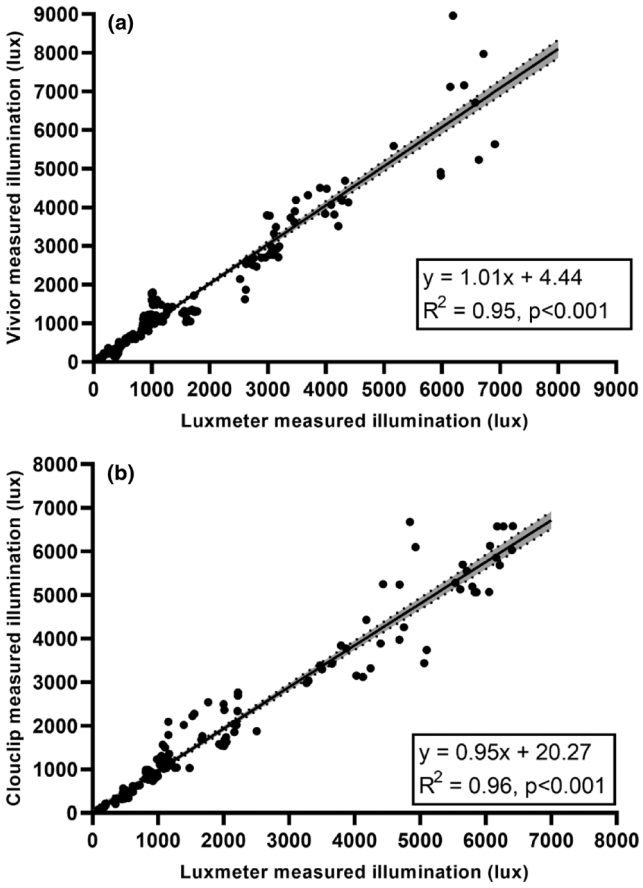


FIGURE 6 Linear regression between the viewing distance captured by (a) the Vivior and luxmeter and (b) the Clouclip and luxmeter. The solid black line represents the 1:1 line and the grey-shaded area indicates the 99% confidence intervals.

eye, and the mean (SEM) best-corrected visual acuity was −0.10 (0.02) logMAR in each eye.

Accuracy in viewing distance measurement

There was no significant difference in viewing distance measured during each near task with the three different measurement methods ($F_{2,108} \leq 1.71$, $p \geq 0.20$; Table 2) under three ambient illuminations (500, 250 and 25 lux). Figure 8 illustrates the mean (SEM) viewing distances during the four near tasks measured with the three different methods under varying illuminations. There was an overall significant difference in viewing distance between the four near-visual tasks ($F_{3,144} \geq 7.90$, $p \leq 0.001$) with a shorter viewing distance when reading a printed article compared to typing on a laptop under 250 lux and 25 lux ambient illuminations measured using all three methods ($p \leq 0.02$ for all) (Figure 8 and Table 3). A significant difference between reading printed articles and typing on a laptop under 500 lux illumination was seen with the Clouclip ($p = 0.02$). The viewing distance while reading an article on an iPad was significantly shorter compared with watching videos on the same iPad under 250 lux when measured with the standard ruler ($p = 0.04$) and under 25 and 500 lux when measured with Clouclip ($p \leq 0.01$ for both).

DISCUSSION

Studies have found prolonged near work with shorter viewing distances[2] and less time spent outdoors[3] to be associated with myopia development. Real-time objective quantification of the viewing distance and exposure to various intensities of ambient illumination is now possible with wearable sensors,[9,13,14] which overcome limitations of recall and parental biases inherent in subjective methods.[6–8] In the current study, the repeatability of viewing distance and ambient illumination measures with the Vivior and Clouclip was investigated. To the best of our knowledge, this is the first study to compare the accuracy of real-time viewing distances measured with the two wearable sensors, Vivior and Clouclip, in human adults while performing different near tasks.

Five Vivior and Clouclip sensors were investigated and found to have good repeatability in measuring viewing distance (testing range: 10–100 cm) across 2 days. The range of mean differences was from −0.11 to +0.15 cm and −0.22 to

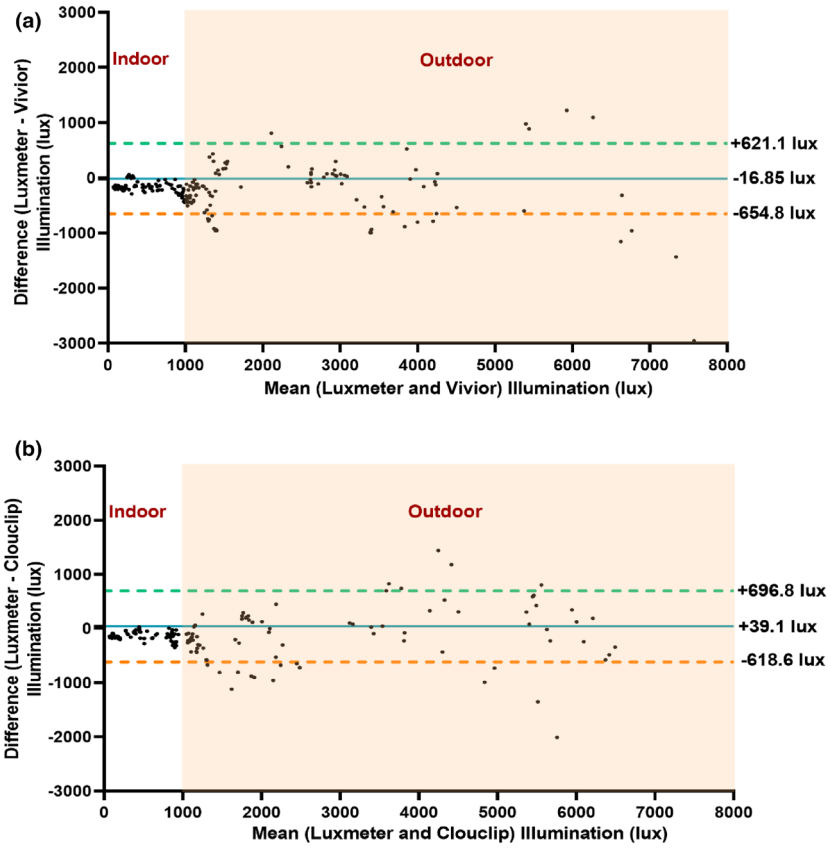


FIGURE 7 Bland–Altman plots of ambient illumination measured at different indoor and outdoor locations for (a) the standard luxmeter and the Vivior and (b) the standard luxmeter and Clouclip. The shaded area represents illumination captured at the outdoor location. The cut-off value >1000 lux for outdoor photopic illumination is based on the previous studies.[18,19] The dashed green and orange lines represent the upper and lower limits of agreement, respectively, and the solid blue line indicates the mean bias.

TABLE 2 Differences in viewing distance during four near-visual tasks measured using the standard ruler, Vivior and Clouclip, under three illuminations.

		Mean (SEM) difference in viewing distance (cm)		
Type of near tasks	**Measurement mode**	**500 Lux**	**250 Lux**	**25 Lux**
Reading printed article	Standard ruler – Vivior	1.03 (0.4) ($p=0.92$)	1.94 (0.6) ($p=0.74$)	1.13 (0.5) ($p=0.91$)
	Standard ruler – Clouclip	−0.64 (0.5) ($p=0.96$)	−1.45 (0.6) ($p=0.84$)	−2.10 (0.6) ($p=0.69$)
	Clouclip – Vivior	1.69 (0.6) ($p=0.79$)	3.39 (0.7) ($p=0.39$)	3.23 (0.9) ($p=0.43$)
Reading an article on an iPad	Standard ruler – Vivior	0.33 (0.5) ($p=0.99$)	−0.75 (0.4) ($p=0.98$)	−0.86 (0.4) ($p=0.97$)
	Standard ruler – Clouclip	−1.16 (0.7) ($p=0.95$)	−1.18 (0.8) ($p=0.95$)	−1.87 (0.5) ($p=0.87$)
	Clouclip – Vivior	1.49 (1.0) ($p=0.91$)	0.43 (0.7) ($p=0.99$)	1.01 (0.7) ($p=0.95$)
Typing on a laptop	Standard ruler – Vivior	0.86 (1.2) ($p=0.98$)	1.98 (0.7) ($p=0.92$)	1.13 (0.7) ($p=0.97$)
	Standard ruler – Clouclip	−1.68 (0.4) ($p=0.94$)	−0.25 (0.7) ($p=0.99$)	−1.35 (0.4) ($p=0.96$)
	Clouclip – Vivior	2.54 (1.1) ($p=0.87$)	2.23 (0.8) ($p=0.89$)	2.48 (0.8) ($p=0.87$)
Watching videos on an iPad	Standard ruler – Vivior	−0.32 (0.9) ($p=0.99$)	0.67 (1.2) ($p=0.99$)	1.35 (0.9) ($p=0.95$)
	Standard ruler – Clouclip	−0.98 (0.9) ($p=0.97$)	−0.68 (0.3) ($p=0.99$)	−1.62 (0.4) ($p=0.92$)
	Clouclip – Vivior	0.66 (1.5) ($p=0.98$)	1.35 (1.2) ($p=0.95$)	2.97 (1.1) ($p=0.77$)

Abbreviations: cm, centimetre; SEM, standard error of mean.

−0.04 cm, respectively. A previous study reported a larger range of differences for viewing distance measurements between 5 and 120 cm taken on 2 days with five Clouclips of −1.0 to +0.6 cm.[11] To assess for accuracy, the current study compared distance measurements with a standard ruler with those taken with the two sensors across a range of

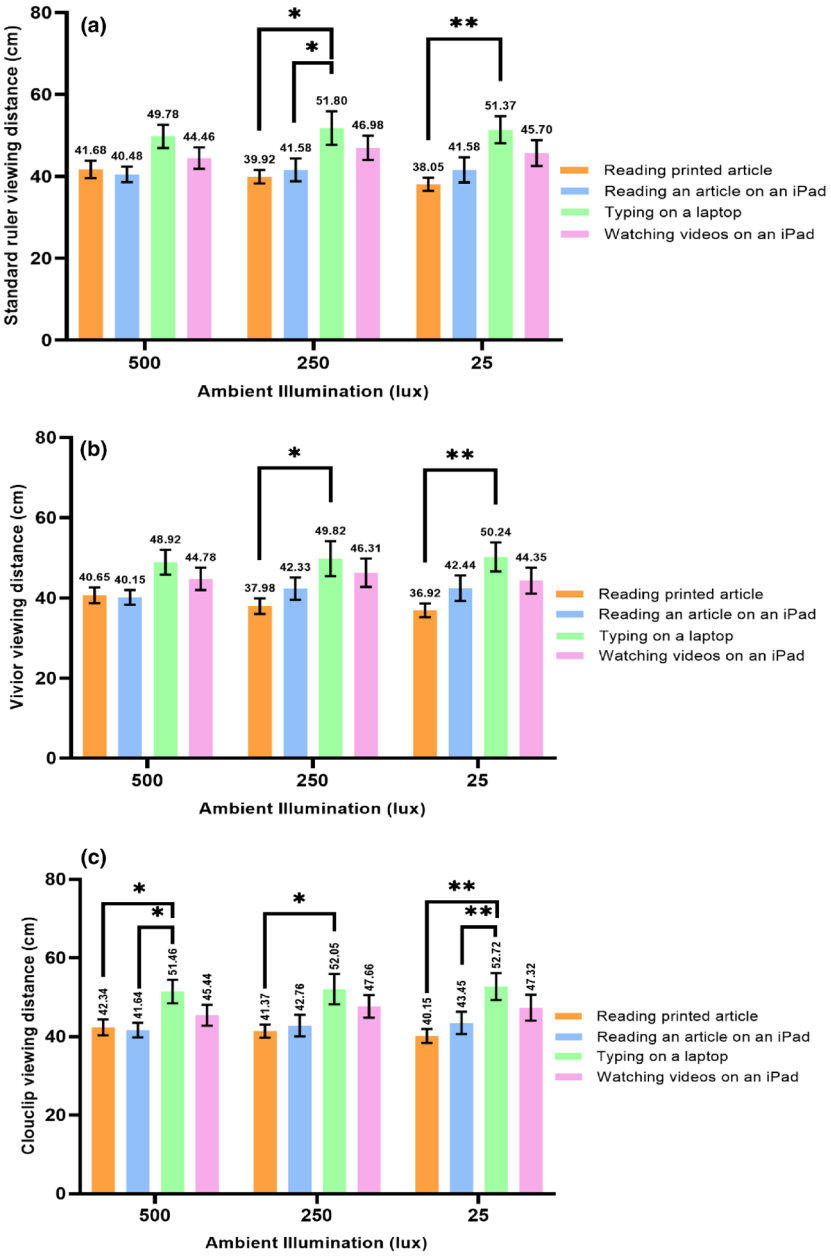


FIGURE 8 Viewing distance measured by the (a) standard ruler, (b) Vivior and (c) Clouclip between four near tasks under three illuminations: 500, 250 and 25 lux. Mean values are presented and error bars represent the standard error of the mean. *Indicates a statistically significant difference.

distances between 10 and 100 cm. Overall, measurements taken with Vivior and Clouclip sensors were comparable to the standard ruler, indicating high accuracy. Mean differences of measurements with the five Vivior and Clouclip devices compared to the standard ruler ranged from −0.22 to +0.11 and −0.27 to +0.01 cm, respectively. Bhandari and Ostrin reported results of 5- to 120-cm viewing distances, with a mean difference between actual distances and the Clouclip of 4.0 and 4.5 cm for a printed novel and an iPad targets, respectively.[11] They also reported a lower mean difference of −0.4 cm for the printed novel and +0.5 cm for the iPad targets for closer testing distances ranging between 15 and 60 cm.[11] The minor discrepancies in results between the current study and that of Bhandari and Ostrin may be attributable to the different distance testing ranges (10–100 cm vs. 5–120 cm) and different viewing targets.

To assess ambient illumination measurements, various indoor and outdoor locations were selected, and overall, Vivior and Clouclip sensors were able to capture illumination levels that were highly correlated with luxmeter measurements. This aligns with Bhandari and Ostrin[11] ($r = 0.96$) and Howell et al.[12] ($r = 0.99$) who also reported a strong correlation between the luxmeter and Clouclip illumination measurements.[11,12] In this study, the mean differences in ambient illumination measurement between the luxmeter and Vivior and luxmeter and Clouclip were −16.85 ± 325.5 and 39.1 ± 335.6 lux, respectively. Howell et al.[12] reported a mean difference of 79.35 lux (LOA −327.98 to +486.68

TABLE 3 Viewing distance difference among the four near-visual tasks captured by standard ruler, Vivior and Clouclip under three illuminations.

Ambient illumination	Type of near-visual task	Mean (SEM) difference in viewing distance (cm)		
		Standard ruler	Vivior	Clouclip
500 Lux	Reading printed article – Reading an article on an iPad	1.20 (1.4) ($p=0.98$)	0.50 (1.0) ($p=0.99$)	0.70 (1.5) ($p=0.97$)
	Reading printed article – Typing on a Laptop	−8.10 (2.4) ($p=0.16$)	−8.27 (2.5) ($p=0.19$)	−9.12 (2.6) **($p=0.02$)***
	Reading printed article – Watching videos on an iPad	−2.78 (2.2) ($p=0.89$)	−4.13 (2.5) ($p=0.75$)	−3.10 (1.9) ($p=0.43$)
	Reading an article on an iPad – Typing on a Laptop	−9.30 (2.2) ($p=0.08$)	−8.77 (2.6) ($p=0.15$)	−9.82 (2.6) **($p=0.01$)***
	Reading an article on an iPad – Watching videos on an iPad	−3.98 (1.5) ($p=0.74$)	−4.63 (2.0) ($p=0.68$)	−3.80 (1.5) ($p=0.10$)
	Typing on a Laptop – Watching videos on an iPad	5.32 (2.1) ($p=0.52$)	4.14 (2.9) ($p=0.75$)	6.02 (2.5) ($p=0.14$)
250 Lux	Reading printed article – Reading an article on an iPad	−1.66 (2.0) ($p=0.97$)	−4.35 (1.8) ($p=0.72$)	−1.39 (1.4) ($p=0.78$)
	Reading printed article – Typing on a Laptop	−11.88 (2.9) **($p=0.01$)***	−11.84 (2.9) **($p=0.02$)***	−10.68 (2.9) **($p=0.01$)***
	Reading printed article – Watching videos on an iPad	−7.06 (2.5) ($p=0.27$)	−8.33 (2.9) ($p=0.19$)	−6.29 (2.2) ($p=0.06$)
	Reading an article on an iPad – Typing on a Laptop	−10.22 (4.2) **($p=0.04$)***	−7.49 (3.8) ($p=0.28$)	−9.29 (3.4) ($p=0.07$)
	Reading an article on an iPad – Watching videos on an iPad	−5.40 (1.6) ($p=0.51$)	−3.98 (2.3) ($p=0.77$)	−4.90 (1.7) ($p=0.06$)
	Typing on a Laptop – Watching videos on an iPad	4.82 (4.3) ($p=0.60$)	3.51 (4.8) ($p=0.83$)	4.39 (3.9) ($p=0.69$)
25 Lux	Reading printed article – Reading an article on an iPad	−3.53 (2.6) ($p=0.80$)	−5.52 (2.5) ($p=0.55$)	−3.30 (2.4) ($p=0.55$)
	Reading printed article – Typing on a Laptop	−13.32 (2.7) **($p=0.003$)****	−13.32 (2.9) **($p=0.001$)****	−12.57 (2.8) **($p=0.004$)****
	Reading printed article – Watching videos on an iPad	−7.65 (2.7) ($p=0.21$)	−7.43 (2.6) ($p=0.28$)	−7.17 (2.7) ($p=0.08$)
	Reading an article on an iPad – Typing on a Laptop	−9.79 (1.9) ($p=0.06$)	−7.80 (2.1) ($p=0.24$)	−9.27 (2.0) **($p=0.003$)****
	Reading an article on an iPad – Watching videos on an iPad	−4.12 (1.6) ($p=0.71$)	−1.91 (1.2) ($p=0.96$)	−3.87 (1.8) ($p=0.19$)
	Typing on a Laptop – Watching videos on an iPad	5.67 (2.2) ($p=0.46$)	5.89 (2.5) ($p=0.49$)	5.40 (2.2) ($p=0.12$)

Abbreviations: cm, centimetre; SEM, standard error of mean.

Note: Bold *p* values indicate statistical significance. *indicates $p<0.05$,**indicates $p<0.005$.

lux) ambient illumination measurement between Clouclip and luxmeter which is slightly higher than observed here. Bhandari and Ostrin[11] found a higher mean difference of 1670.4 lux (LOA −6289 to +9629.8 lux), likely due to the comparatively larger illumination range (up to 30,000 lux) tested.

Variability of measurements between Vivior and Clouclip compared to the luxmeter increased with outdoor illumination levels >3500 lux. Similarly, Howell et al.[12] reported greater variation between the luxmeter and Clouclip during outdoor illumination >2500 lux. Although both wearable sensors accurately captured indoor photopic illumination levels (>3 and <1000 lux), they underestimated outdoor photopic illumination levels (>1000 lux[12,19]) which may be due to the highly variable nature of outdoor lighting conditions. Despite increased variability and reduced accuracy of measurements taken at illumination levels >3500 lux, as outdoor photopic conditions are typically considered to be ≥1000 lux light intensity,[13,19] both Vivior and Clouclip sensors were able to capture changes from indoor to outdoor (or vice versa) environments. It is important that the variability in capturing outdoor illumination reported in this study and previously reported by Howell et al.[12] for Clouclip be considered in future clinical studies utilising these sensors to investigate the impact of the intensity of outdoor illumination on myopia progression, as increased variability has been noted in high illumination

environments. It is worth noting that, in current and previous studies,[11,12] wearable sensors captured illumination measurements from a smaller sampling area compared to standard lux meters, which may be responsible for some measurement discrepancy between methods. Wearable sensors are designed to measure illumination from a limited range of the central visual field; in the current study, the illumination was captured with the sensors directed at primary gaze. Therefore, illumination captured through the mid-peripheral to the peripheral portion of the luxmeter sensor may impact the measurements. In this study, the Vivior, Clouclip and luxmeter recorded >1000 lux illumination for some indoor locations, such as near a window or glass door on a bright sunny day. This aligns with the outcomes of a previous study validating ambient illumination using the Actiwatch (https://www.philips.com.gh/healthcare/product/HC1044809/actiwatch-2-activity-monitor) and Clouclip, which reported >1000 lux illumination for some indoor locations, such as near a window or door with bright sunlight.[12] The sensitivity and specificity values were acceptable for differentiating indoor and outdoor locations for Vivior and Indoors.

Study 2 compared the accuracy of viewing distance measurements during four near tasks under three different illumination levels while simultaneously using the Vivior, Clouclip and a standard ruler. There were no significant differences in mean viewing distances captured by the three methods during any of the four near-visual tasks, indicating high accuracy of both devices for real-time quantification of near-viewing distance supporting the results of Study 1. Shorter working distances were measured with all devices when reading a printed article compared to typing on a laptop, as well as reading an article compared to watching a video on the iPad (Figure 8). Bhandari and Ostrin also reported a significantly shorter viewing distance with the Clouclip during a near task with a paper target compared with the iPad target.[11] The results indicated that paper-based targets and near tasks involving reading may encourage shorter working distances. The reason for the minor, non-statistically significant discrepancies in viewing distances between the different methods is unknown, but all measurements showed similar trends (Figure 8 and Table 3). For example, at 25 lux luminance, all methods showed a slightly shorter viewing distance (<2 cm on average) when reading an article on an iPad compared to typing on a laptop, but this trend was significant only for the Clouclip. A study power of 0.83 with an alpha error probability of 0.05 was calculated (G*Power 3.1.9.7), indicating that the current sample size was adequate to support the findings.

The Clouclip has previously been used to capture near viewing distance and duration, as well as light exposure in myopic children[9,13,14] and adults.[15] However, this is the first study to report the accuracy of a new wearable sensor, the Vivior, in capturing viewing distance during near tasks in adults. In the literature, apart from the Clouclip and Vivior, another wearable sensor, the Rangelife (adafruit.com/product/3317),[20] has also been explored and reported as a reliable device to capture habitual near viewing activity in myopic and non-myopic adults. Recently, a new wearable sensor, the Visual Environment Evaluation Tool (VEET) (projectveet.com), has also been validated for capturing viewing distances using a mannequin head but not evaluated in humans.[21]

Some technical limitations were experienced while conducting the study. The Clouclip was found to go into sleep mode if no movement was detected for more than 40 s, which has been observed previously.[11] Despite tilting spectacle frames every 40 s, there were missing data from the Clouclip of approximately 1–2 min in both Study 1 and Study 2. If there is no significant head movement during near activities, data may be lost with the Clouclip. Additionally, during near activities in Study 2, the Clouclip was tilted an additional 10° downwards[11] to support the capture of viewing distances when participants did not tilt their head downwards and only changed their eye gaze. This was not required for the Vivior as it includes two sensors, one of which is positioned 30° downwards to detect the near target for measurements when there is no head movement downwards.

CONCLUSIONS

Both Vivior and Clouclip sensors showed excellent repeatability and accuracy in measuring real-time viewing distance and ambient illumination in a controlled laboratory setting with a mannequin head. When used by young adult participants, Vivior and Clouclip were able to capture reliable viewing distances accurately during four common near tasks, all comparable to standard ruler measurements. Although not statistically significant, the repeatability in viewing distance measurements varied slightly among the five Vivior and Clouclip sensors. Ambient illumination measurements were also highly repeatable with the Vivior and Clouclip, although measurements were more variable and less than that measured with the lux meter for levels >3500 lux. However, as outdoor illumination levels are typically ≥1000 lux,[13,19] and considering both devices were able to demonstrate high accuracy and low variability within this illumination range, the Vivior and Clouclip can accurately detect whether the subject is in an indoor or outdoor environment. The Bland–Altman analysis conducted in Study 1 showed overall high agreement between the standard ruler and Vivior or Clouclip, as no significant bias was found across all the tested distances (Figures 4 and 5 and Table 1). In study 2, the mean differences in viewing distance between Clouclip and Vivior during all four near

tasks under three ambient illuminations were also not significant (Table 2), indicating that the two sensors are interchangeable for real-time viewing distance and ambient illumination measurements. Therefore, Vivior and Clouclip sensors are recommended for real-time capture of habitual near viewing duration, distance and light exposure in any future study that aims to evaluate the effect of visual environment on any ocular disease.[22] However, limitations of the sensors such as sleep mode, extra tilt to capture viewing distance more accurately, compliance of spectacle use for non-spectacle users (e.g., contact lens wearers and emmetropic individuals) must be considered.

AUTHOR CONTRIBUTIONS

Samrat Sarkar: Conceptualization (equal); data curation (lead); formal analysis (lead); investigation (lead); methodology (equal); project administration (lead); writing – original draft (lead); writing – review and editing (equal). **Sieu K. Khuu:** Conceptualization (equal); data curation (equal); methodology (equal); supervision (equal); validation (equal); writing – review and editing (equal). **Fuensanta A. Vera-Diaz:** Conceptualization (equal); data curation (equal); funding acquisition (equal); methodology (equal); supervision (equal); validation (equal); writing – review and editing (equal). **Pauline Kang:** Conceptualization (equal); data curation (equal); funding acquisition (lead); methodology (equal); resources (lead); supervision (lead); validation (equal); writing – review and editing (equal).

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CONFLICT OF INTEREST STATEMENT

None.

DATA AVAILABILITY STATEMENT

The data are available from the authors upon valid request from the readers.

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